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Recent Enhancements to the TC Police Action

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Agenda

- Packet-per-second rate-limiting
- Modeling OpenFlow Metering with Policer Action



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Packet-Per-Second Rate-Limiting

Packet-per-Second Rate-Limiting

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PPS

- Add mechanism to apply rate-limit based on Packets/s
- Now a policer action instance may enforce rate limit as either:
  - Bits/s (long-standing behaviour)
  - Packets/s (new behaviour)
- Modes may be combined using multiple actions

```
tc filter add dev $dev ingress protocol ip \
    flower ip_proto udp dst_port 54321 \
    action police pkts_rate 2000 pkts_burst 400 \
    action police rate 24Mbit burst 600Kb
```

Hardware Offload of PPS

- Offload via Flow Rule API
- Support dependent on offload driver
- Currently only supported by NFP driver with flower

```
struct flow_action_entry {
    ...
    struct { /* FLOW_ACTION_POLICE */
        u32  index;
        u32  burst;
        u64  rate_bytes_ps;
        u64  burst_pkt;
        u64  rate_pkt_ps;
        u32  mtu;
    } police;
    …
}
```

Status of PPS

Packet-per-second rate limiting is fully upstream

- Kernel
  - v4.13
  - TC policer action
  - Flow rule
  - NFP driver
- OVS
  - v2.16, pending release
  - port-based ingress rate limiting configured via OVSDB

Open Flow Meters meet the TC Police Action

Open Flow Meters

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Open Flow Meters

- n-band

- Each band has a target rate
  - bytes/s or packets/s
- Drop or DSCP mark packets

- Are independent of flows
  - May be used by zero or more flows via meter action
- Are independent of ports
  - May be used in flows attached to different ports

Meter Example

```
ovs-ofctl -O openflow13 add-meter br0  \
    meter=99,kbps,burst,band=type=drop,rate=1000,burst_size=100

for dev in $DEV1 $DEV2; do
    ovs-ofctl -O openflow13 add-flow br0 \
        in_port=$dev,ip,action=meter:99,output=$DEV3 \
    ovs-ofctl -O openflow13 add-flow br0 \
        in_port=$dev,ipv6,action=meter:99,output=$DEV3 \
done
```

Meters and Police Action

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Open Flow Meters

- n-band
- Each band has a target rate
  - bytes/s or packets/s
- Drop or DSCP mark packets
- Are independent of flows
  - May be used by zero or more flows via meter action
- Are independent of ports
  - May be used in flows attached to different ports

TC Police Action

- 1-band
  - multiple bands via multiple actions
- Each instance has a target rate
  - bytes/s or packets/s
- May drop packets
  - Or mark with assistance from other actions
- May be independent of filters
  - May be used by zero or more flows using index of action instance
- May be independent of netdevs
  - May be used by filters attached to different ports

Wow, that's a pretty good match!

TC Action Independent of Filters

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```
tc action add action police index 99 rate 1mbit burst 100k

for dev in $DEV1 $DEV2; do
    tc qdisc add dev $dev ingress
    tc filter add dev $dev ingress protocol ip \
        flower ip_proto tcp action police index 99
    tc filter add dev $dev ingress protocol ipv6 \
        flower ip_proto tcp action police index 99
done
```

Hardware Offload Prototype

Offload via Flow Rule API

- New tc_setup_type
  - TC_SETUP_ACT
  - First class citizen along with U32, Flower, Block, ...
- Action setup via:
  - cls_api.c: tc_setup_flow_action()
  - cls_api.c: tcf_action_init()
- Action deletion via:
  - tcf_action_offload_cmd()
- Action statistics via:
  - tcf_exts_stats_update()

Status of Meters via TC

Packet-per-second rate limiting is fully upstream

- Kernel
  - RFC Posted
  - Targeting inclusion in v5.15
  - Flow rule
  - NFP driver
- OVS
  - Targeting v2.17
  - Meters are configured via Open Flow


THANKS